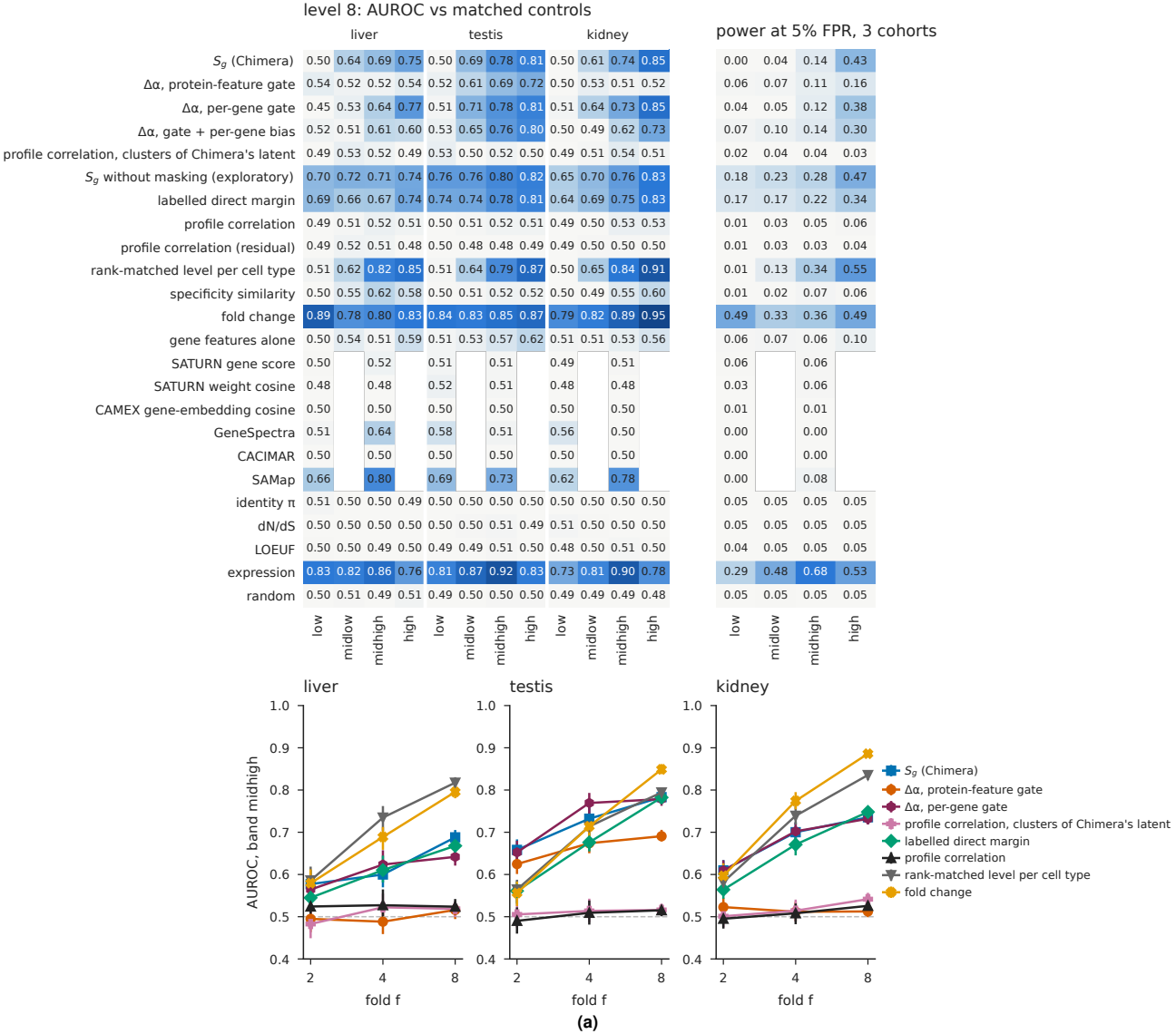
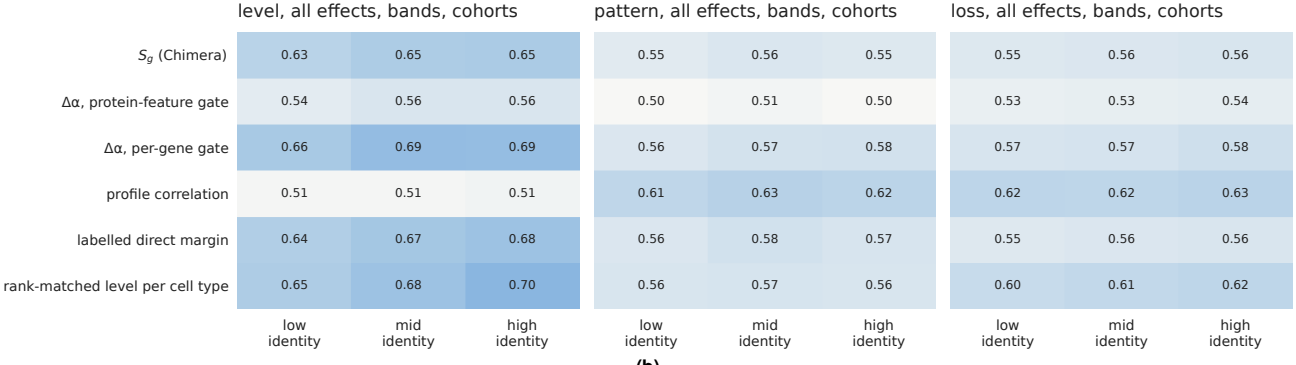

level 8: AUROC vs matched controls
liver
testis
kidney
power at 5% FPR, 3 cohorts
S_g (Chimera) 0.50 0.64 0.69 0.75 0.50 0.69 0.78 0.81 0.50 0.61 0.74 0.85 0.00 0.04 0.14 0.43
Δα, protein-feature gate 0.54 0.52 0.52 0.54 0.52 0.61 0.69 0.72 0.50 0.53 0.51 0.52 0.06 0.07 0.11 0.16
Δα, per-gene gate 0.45 0.53 0.64 0.77 0.51 0.71 0.78 0.81 0.51 0.64 0.73 0.85 0.04 0.05 0.12 0.38
Δα, gate + per-gene bias 0.52 0.51 0.61 0.60 0.53 0.65 0.76 0.80 0.50 0.49 0.62 0.73 0.07 0.10 0.14 0.30
profile correlation, clusters of Chimera's latent 0.49 0.53 0.52 0.49 0.53 0.50 0.52 0.50 0.49 0.51 0.54 0.51 0.02 0.04 0.04 0.03
S_g without masking (exploratory) 0.70 0.72 0.71 0.74 0.76 0.76 0.80 0.82 0.65 0.70 0.76 0.83 0.18 0.23 0.28 0.47
labelled direct margin 0.69 0.66 0.67 0.74 0.74 0.74 0.78 0.81 0.64 0.69 0.75 0.83 0.17 0.17 0.22 0.34
profile correlation 0.49 0.51 0.52 0.51 0.50 0.51 0.52 0.51 0.49 0.50 0.53 0.53 0.01 0.03 0.05 0.06
profile correlation (residual) 0.49 0.52 0.51 0.48 0.50 0.48 0.48 0.49 0.49 0.50 0.50 0.50 0.01 0.03 0.03 0.04
rank-matched level per cell type 0.51 0.62 0.82 0.85 0.51 0.64 0.79 0.87 0.50 0.65 0.84 0.91 0.01 0.13 0.34 0.55
specificity similarity 0.50 0.55 0.62 0.58 0.50 0.51 0.52 0.52 0.50 0.49 0.55 0.60 0.01 0.02 0.07 0.06
fold change 0.89 0.78 0.80 0.83 0.84 0.83 0.85 0.87 0.79 0.82 0.89 0.95 0.49 0.33 0.36 0.49
gene features alone 0.50 0.54 0.51 0.59 0.51 0.53 0.57 0.62 0.51 0.51 0.53 0.56 0.06 0.07 0.06 0.10
SATURN gene score 0.50 0.52 0.51 0.51 0.49 0.51 0.06 0.06
SATURN weight cosine 0.48 0.48 0.52 0.51 0.48 0.48 0.03 0.06
CAMEX gene-embedding cosine 0.50 0.50 0.50 0.50 0.50 0.50 0.01 0.01
GeneSpectra 0.51 0.64 0.58 0.51 0.56 0.50 0.00 0.00
CACIMAR 0.50 0.50 0.50 0.50 0.50 0.50 0.00 0.00
SAMap 0.66 0.80 0.69 0.73 0.62 0.78 0.00 0.08
identity π 0.51 0.50 0.50 0.49 0.50 0.50 0.50 0.50 0.50 0.50 0.50 0.50 0.05 0.05 0.05 0.05
dN/dS 0.50 0.50 0.50 0.50 0.50 0.50 0.51 0.49 0.51 0.50 0.50 0.50 0.05 0.05 0.05 0.05
LOEUF 0.50 0.50 0.49 0.50 0.49 0.49 0.51 0.50 0.48 0.50 0.51 0.50 0.04 0.05 0.05 0.05
expression 0.83 0.82 0.86 0.76 0.81 0.87 0.92 0.83 0.73 0.81 0.90 0.78 0.29 0.48 0.68 0.53
random 0.50 0.51 0.49 0.51 0.49 0.50 0.50 0.50 0.49 0.49 0.49 0.48 0.05 0.05 0.05 0.05
low midlow midhigh high
liver
testis
kidney
AUROC, band midhigh
fold f
S_g (Chimera)
Δα, protein-feature gate
Δα, per-gene gate
profile correlation, clusters of Chimera's latent
labelled direct margin
profile correlation
rank-matched level per cell type
fold change

(a)


level, all effects, bands, cohorts
pattern, all effects, bands, cohorts
loss, all effects, bands, cohorts
S_g (Chimera) 0.63 0.65 0.65 0.55 0.56 0.55 0.55 0.56 0.56
Δα, protein-feature gate 0.54 0.56 0.56 0.50 0.51 0.50 0.53 0.53 0.54
Δα, per-gene gate 0.66 0.69 0.69 0.56 0.57 0.58 0.57 0.57 0.58
profile correlation 0.51 0.51 0.51 0.61 0.63 0.62 0.62 0.62 0.63
labelled direct margin 0.64 0.67 0.68 0.56 0.58 0.57 0.55 0.56 0.56
rank-matched level per cell type 0.65 0.68 0.70 0.56 0.57 0.56 0.60 0.61 0.62
low identity
mid identity
high identity

(b)